

1. Administrative & Study Identification

Full Study Title Phase I/II, Open Label, Multicenter Study of Rapcabtagene Autoleucel in Adult Patients With CLL/SLL, 3L+ DLBCL, r/r ALL and 1L HR LBCL **Short Title/Acronym** Phase I/II Study of Rapcabtagene Autoleucel in CLL, 3L+ DLBCL, r/r ALL and 1L HR LBCL **Protocol Number** CYTB323A12101 **NCT Number** NCT03960840 **Posting ID** 624142759273622627 **Trial Status** Active, Not Recruiting **Sponsor/Company** Novartis Pharmaceuticals Corporation **Investigational Product** Rapcabtagene autoleucel (YTB323, a CD19-directed CAR-T cell therapy manufactured via the T-Charge platform) and Ibrutinib **Disease Areas** Chronic Lymphocytic Leukemia, Small Lymphocytic Lymphoma, Diffuse Large B-cell Lymphoma, Acute Lymphoblastic Leukemia, Large B-cell Lymphoma **Phase of Development** Phase 1 / Phase 2 **Medical Monitor** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** In Phase I, to evaluate the feasibility, dose recommendation (identify dose-limiting toxicities [DLTs]), characterize safety (incidence and severity of AEs/SAEs, labs, ECG, and vital signs), and tolerability (ibrutinib dose modifications in the CLL/SLL arm). In Phase II, to evaluate preliminary antitumor efficacy. * **Secondary Objectives:** To evaluate Complete Response (CR)/Partial Response (PR) in CLL/SLL, Best Overall Response (BOR) of CR/PR per Lugano criteria in 3L+ DLBCL, Duration of Response (DOR) in CLL/SLL and 3L+ DLBCL, Progression-Free Survival (PFS), overall safety, and cellular kinetics (e.g., Cmax).

2.2 Background & Rationale Despite novel treatment options, a substantial portion of patients with relapsed/refractory B-cell malignancies (such as diffuse large B-cell lymphoma, CLL, and ALL) do not achieve sustained remission. Furthermore, the traditional CAR-T manufacturing process can take several weeks. Rapcabtagene autoleucel utilizes the novel T-Charge™ platform to rapidly manufacture CAR-T cells in less than 2 days, preserving T-cell stemness, reducing the "vein-to-door" time for patients, and potentially leading to better expansion, persistence, and deeper clinical responses.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (three independent treatment arms in Phase I; two

independent cohorts in Phase II) * **Blinding:** Open-label * **Randomization:** N/A (Dose escalation followed by dose expansion cohorts)

3.2 Planned Enrollment & Duration * Targeted Enrollment (\$N\$): 216 participants * **Cohort Breakdown:** Approximately 100 participants for the 3L+ DLBCL cohort and 50-60 participants for the 1L HR LBCL cohort planned for primary efficacy analysis. The Phase I CLL/SLL arm completed enrollment as of 05-May-2021. * **Number of Sites:** Multicenter. * **Estimated Study Start:** 2019 * **Estimated Primary Completion:** Ongoing long-term follow-up

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients with diagnosis of Chronic Lymphocytic Leukemia (CLL) or Small Lymphocytic Lymphoma (SLL) according to iwCLL criteria; Relapsed/Refractory Diffuse Large B-cell Lymphoma ($\geq 3L$ DLBCL) after 2 or more lines of therapy including autologous HSCT; Refractory or relapsed CD19-positive Acute Lymphoblastic Leukemia (ALL); or First-Line High-Risk Large B-Cell Lymphoma (1L HR LBCL). 2. Performance Status: ECOG performance status 0-1 for ALL and DLBCL; ECOG performance status 0-2 for 1L HR LBCL at screening. 3. For ALL: Must have morphologic disease in the bone marrow. 4. For CLL/SLL: Must be in SD or PR after at least 6 months of ibrutinib, either as second or subsequent line of therapy. 5. For 1L HR LBCL: Considered high-risk based on IPI score of 3-5 at diagnosis, or MYC and BCL2 and/or BCL6 rearrangement (DH/TH lymphoma). 6. For 1L HR LBCL: Participants must have received 2 cycles of frontline therapy (e.g., R-CHOP, Pola-R-CHP, or DA-EPOCH-R). Participants with DH/TH lymphoma must have received at least one cycle (the most recent) DA-EPOCH-R. 7. For 1L HR LBCL: Must have a positive PET per Lugano classification (Deauville PET score of 4 or 5 and an overall response of PR/SD) after 2 cycles of frontline CIT.

4.2 Exclusion Criteria 1. Prior CD19-directed therapy or prior administration of a genetically engineered cellular product. 2. Prior allogeneic hematopoietic stem cell transplantation (HSCT). 3. Active central nervous system (CNS) lymphoma or prior history of CNS involvement by malignancy (for 1L HR LBCL), primary DLBCL of the CNS, Richter's transformation, or Burkitt lymphoma. 4. Additional exclusions for 1L HR LBCL: DLBCL associated with chronic inflammation, intravascular large B-cell lymphoma, ALK-positive large B-cell lymphoma, HHV8 positive LBCL, DLBCL leg type, or EBV positive DLBCL, NOS. 5. Targeted small molecule or kinase inhibitor within 2 weeks from leukapheresis (unless specified by the protocol like ibrutinib for CLL).

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Single infusion of rapcabtagene autoleucel. Evaluated as a single agent in DLBCL, ALL, and 1L HR LBCL. In CLL/SLL, it is given in combination with daily ibrutinib. *** Route of Administration:** Intravenous (IV) infusion (following lymphodepletion). *** Duration of Treatment:** Single CAR-T cell infusion. Patients are followed for a minimum of 2 years on the main protocol, then roll over into a 15-year long-term safety follow-up for lentiviral vector safety.

5.2 Concomitant & Prohibited Therapy * Permitted: Optional bridging chemotherapy prior to infusion; Lymphodepleting chemotherapy. *** Prohibited:** Targeted small molecule or kinase inhibitors within 2 weeks from leukapheresis.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Eligibility Assessment, Leukapheresis
Baseline	Day -5 to 0	Lymphodepleting chemotherapy, followed by CAR-T Infusion
Interim	Months 1, 3, 6, 12	Safety monitoring (CRS/ICANS), Efficacy Assessment (PET/CT), Cellular Kinetics (qPCR)
End of Study	Year 2	Final study assessments; transfer to 15-year Long-Term Follow-Up (LTFU) destination protocol

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * Phase 1 Dose recommendation: Incidence and nature of Dose Limiting Toxicities (DLTs) in the dose escalation part. *** Phase 1 Safety:** Incidence and severity of AEs and SAEs, including changes in laboratory values, ECG, and vital signs. *** Phase 1 Tolerability:** Ibrutinib dose modifications in the CLL/SLL arm.

7.2 Secondary & Exploratory Endpoints * Phase 1 Efficacy: Complete Response (CR) / Partial Response (PR) in CLL/SLL; Best Overall Response (BOR) of CR/PR per Lugano criteria in 3L+ DLBCL. *** Phase 1 Durability:** Duration of response (DOR) in CLL/SLL and 3L+ DLBCL. *** Progression-Free Survival (PFS)** *** Cellular Kinetics** (Cmax and AUC by qPCR)

8. Safety & Adverse Event Reporting

- Adverse Event (AE) Monitoring:** Special focus on immune-mediated toxicities typical of CAR-T therapies, including Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS), as well as changes in labs, ECGs, and vital signs.

- **Serious Adverse Events (SAEs):** Routine monitoring for grade 3/4 events like prolonged pancytopenia, severe CRS, or hemophagocytic lymphohistiocytosis.
 - **Data Monitoring Committee (DMC):** Independent safety review committee to govern dose-escalation cohorts and expansions.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Efficacy Analysis Set (patients treated with the recommended dose).
 - **Sample Size Justification:** A total targeted enrollment of 216 is expected. Approximately 100 participants in the 3L+ DLBCL cohort (including those from Phase I treated at the RD) and 50-60 in the 1L HR LBCL cohort are planned for the primary efficacy analysis.
 - **Handling of Missing Data:** Standard censoring methods for time-to-event endpoints (PFS, DOR).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** High-frequency onsite monitoring in the first 28 days post-infusion for acute toxicities.
 - **Audit Trail:** Electronic Data Capture (EDC) systems utilized with validated audit trails and source data verification.
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11. Study Identifiers & Governance

- **Primary ID:** CYTB323A12101
 - **Registry Number:** NCT03960840
 - **Posting ID:** 624142759273622627
 - **Sponsor/Lead Organization:** Novartis Pharmaceuticals Corporation
 - **Study Title:** Phase I/II Study of Rapcabtagene Autoleucel in CLL, 3L+ DLBCL, r/r ALL and 1L HR LBCL
 - **Official Regulatory Title:** Phase I/II, Open Label, Multicenter Study of Rapcabtagene Autoleucel in Adult Patients With CLL/SLL, 3L+ DLBCL, r/r ALL and 1L HR LBCL
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12. Clinical Framework (CAR-T Specific)

- **Primary Conditions:** Chronic Lymphocytic Leukemia, Small Lymphocytic Lymphoma, Diffuse Large B-cell Lymphoma, Acute Lymphoblastic Leukemia, Large B-cell Lymphoma
- **Diagnosis Threshold:** Confirmed specific histologies (ALL requiring morphologic disease in bone marrow; 1L HR LBCL

requiring IPI 3-5 or DH/TH and Deauville PET score 4-5 PR/SD).

- **Medical Methodology:** Next-generation Autologous CD19-directed CAR-T Cell Therapy (T-Charge™ platform)
 - **Optimization Target:** Stem-like memory T-cell preservation, maximizing in vivo expansion, and achieving deep clinical responses.
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13. Study Procedures & Methodology

- **Management Pathway:** Patient leukapheresis → Rapid manufacturing (<2 days) → Bridging therapy (optional) → Lymphodepletion → Infusion (with Ibrutinib for CLL/SLL arm).
 - **Education Protocol (HCP):** Specialized cellular therapy center training for administering and managing CAR-T toxicities (CRS/ICANS).
 - **Education Protocol (Patient):** Counseling regarding the requirements of cellular therapy, risk of infections, and prolonged monitoring.
 - **Quality Audit Mechanism:** Vector tracking, centralized lab evaluation of cellular kinetics (qPCR), and 15-year post-study lentiviral vector safety tracking.
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14. Observation & Follow-Up Schedule

- **Total Duration:** 2 years on protocol, transitioning to a 15-year follow-up destination protocol for lentiviral vector safety.
 - **Onsite Visit Cadence:** Intensive monitoring (often inpatient or near-site outpatient) during the first 28 days post-infusion, shifting to quarterly and annual visits.
 - **Remote Monitoring Frequency:** As per institutional standards for long-term survivors.
 - **Checklist Review Interval:** Regular tracking for progressive disease or secondary malignancies during the long-term phase.
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				